

# Repository-wise C++ code to Java Translation leveraging the LLM

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## ABSTRACT

Code translation—converting code from one programming language to another while preserving functionality—is critical for legacy system migration, cross-platform development, and ecosystem modernization. Traditional rule-based or manual approaches are limited in scope and efficiency. While large language models (LLMs) have shown promise in code generation, most existing research focuses on translating isolated methods or small files, leaving project-level translation—where accumulated language paradigm differences become significant—largely unexplored.

This paper proposes TOOL, a hierarchical, macro-to-micro translation framework designed to mitigate error propagation in large-scale code translation. The framework first performs static analysis to extract call graphs and decompose code into functional modules using a custom Java parser tool. Translation then proceeds in two layers: (1) at the class level, where overall architecture and header files are designed with language-specific pattern mapping, and (2) at the method level, where implementations are translated bottom-up along call chains to ensure dependency stability. To bridge library and runtime differences, the framework dynamically generates necessary dependencies and an autonomous agent iteratively compiles, tests, and repairs the translated code using intelligent error analysis and fix strategies.

We evaluate TOOL by translating several cohesive Java modules to C++, two languages with significant differences in underlying execution models. Experimental results demonstrate that ... **TODO: Add experimental results**

Keywords: Code translation, Large language models, Project-level translation, Java, C++, Static analysis, Hierarchical translation

## 1 INTRODUCTION

The increasing need for software migration, cross-language interoperability, and multi-platform deployment has made automated code translation a valuable research area. Early techniques relied on handcrafted rules or template-based transformations, which are difficult to scale. Recent advances in LLMs have enabled more flexible and context-aware code generation, yet their application in translation remains largely confined to small, self-contained code snippets. Translating entire projects or interconnected modules introduces challenges such as architectural mismatch, dependency management, and compounding semantic gaps between languages—especially those with different runtime paradigms like Java (managed, JVM-based) and C++ (unmanaged, native).

This paper addresses the problem of error accumulation in project-level translation. We argue that translating files or methods independently without a high-level understanding of

the module’s purpose leads to irreconcilable inconsistencies. Instead, we propose a top-down understanding followed by bottom-up translation, where the LLM first comprehends the macro-structure of a functional module, plans the overall translation strategy, and then executes translation in a layered manner.

Our main contributions are:

- **A hierarchical translation framework (tool)** that separates class-level design from method-level implementation, enabling macro-to-micro translation with reduced error propagation.
- **A call-graph-guided, bottom-up translation order** that stabilizes dependencies by translating leaf methods first, ensuring callees are available when translating callers.
- **An autonomous compilation and repair agent** that validates and corrects the translated code using intelligent error analysis, supporting iterative compilation with automated fix strategies.

## 2 RELATED WORK

### 2.1 Traditional Code Translation Methods

Early work recognized that programs possess rich syntactic structures and that direct application of sequence-to-sequence models often yields syntactically invalid outputs. To address this, Chen et al. [1] pioneered the use of tree-to-tree neural networks, which encode and decode abstract syntax trees (ASTs) to explicitly model program structure. This was further refined by Drissi et al. [3], who introduced a grammar-driven tree-to-tree model that hard-codes target language syntax rules into the decoder, guaranteeing grammatical correctness. These studies established the importance of structural awareness in code translation but were limited by their dependence on parsers, manually defined grammars, and parallel tree-aligned corpora, which hindered generalization and scalability.

To overcome the scarcity of parallel data, The field shifted toward data-driven, unsupervised methods. Roziere et al. [17] introduced TransCoder, which leverages back-translation and denoising auto-encoding on monolingual code corpora to learn cross-lingual code representations without supervision. Concurrently, pre-trained models such as CodeBERT and CodeT5 adapted the pre-training paradigm from natural language processing to code, learning general-purpose representations via masked language modeling. These approaches demonstrated the power of scale and transfer learning, moving the focus from designing task-specific architectures to fine-tuning general-purpose models. However, Malyala et al. [11] found that purely data-driven models often exhibit systematic errors in complex translation scenarios, such as type

conversions and loop restructuring. Pan et al.[14] conducted a large-scale empirical study showing that LLMs introduce a significant number of bugs during translation, particularly in real-world projects, highlighting the limitations of statistical approaches in understanding deep semantic and syntactic constraints.

## 2.2 LLM Based Code Translation Methods

To mitigate the shortcomings of purely data-driven models, recent research emphasizes augmenting LLMs with programming language-specific knowledge, such as type systems, safety rules, and API mappings. This is especially critical for security-sensitive migrations, such as from C to Rust. Luo et al. [10] proposed the IRENE framework, which combines static analysis to extract “high-risk patterns” with retrieval-augmented generation to guide the model toward producing Rust code that complies with ownership rules. Similarly, Hong et al. [4] focused on type migration, employing a call-graph-based strategy to coordinate interface changes across functions. These hybrid systems effectively address the “semantic drift” and safety violations common in naive LLM translations. Extending this neuro-symbolic approach, Ibrahimzada et al. [5] introduced AlphaTrans, which compositionally combines symbolic knowledge with neural models for repository-level translation and validation. Meanwhile, Nitin et al.[12] enhances translation by generating multi-modal specifications (e.g., comments, tests) to provide additional context. While these methods significantly improve reliability, they often require substantial engineering effort to construct and maintain domain-specific knowledge bases.

## 2.3 Hallucination in General-Purpose LLMs

Hallucination in large language models (LLMs) refers to the generation of factually incorrect or contextually inconsistent content. Early research focused on text-based hallucinations, with recent work expanding to multimodal settings [15]. Detection methods have evolved from lexical metrics (ROUGE, BLEU) to more robust frameworks incorporating multi-dimensional evaluation [2] and domain-specific approaches like self-reflection in medical QA [6]. Mitigation strategies include knowledge augmentation, prompt engineering, and adversarial defenses [16, 19].

Code generation introduces unique challenges due to strict syntactic and functional requirements. Liu et al. [8] identified three primary hallucination types: Requirement Conflict, Knowledge Hallucination, and Code Inconsistency. In repository-level generation, Zhang et al.[22] further categorized these into Task Requirement Conflicts, Factual Knowledge Conflicts, and Project Context Conflicts.

The causes are multifaceted, including training data quality issues, intent comprehension limitations, and inadequate context awareness [22]. Impacts are severe: 95.30% of hallucinations cause functional errors, with hallucinated code having only 6.26% pass@1 rate versus 54.73% for clean code [8].

Retrieval-augmented generation (RAG) has proven effective, improving pass@1 scores across models (e.g., +4.38% for ChatGPT) [22]. Other approaches include self-refinement prompts, chain-of-thought reasoning, and specialized code models.

## 2.4 Repo-level Code Translation Methods

As model capabilities have grown, the research scope has expanded from function-level snippets to more complex, practical scenarios. Repository-level code translation presents unique challenges, including managing inter-file dependencies, build configurations, and project-wide context. Wang et al. [18] introduced RepoTransBench, a large-scale, multi-language benchmark for evaluating repository-level translation, revealing that even state-of-the-art LLMs struggle (with success rates around 32.8%). To tackle this, Zhang et al. [21] proposed a skeleton-guided translation framework that first extracts a high-level structural skeleton in the target language before filling in implementation details. Yuan et al.[20] and Ou et al.[13] further explored project-level translation by synergistically integrating knowledge graphs and LLMs, or by evolving a triple knowledge-augmented LLM within the repository context.

Parallel to scaling up, improving the human-in-the-loop experience has become a key focus. Liu et al.[9] developed hmCodeTrans, an interactive translation framework that allows developers to provide real-time feedback. By employing attention caching and suffix splicing, the system achieves millisecond-level responsiveness, reducing human effort by approximately 65% compared to traditional post-editing workflows. This marks a shift from pursuing full automation toward designing collaborative tools that integrate seamlessly into developer workflows.

Our work is situated at this intersection. We aim to develop a translation framework that leverages the generative strength of LLMs while deeply integrating extensible, domain-aware knowledge and validation mechanisms. We target translation tasks involving complex algorithm logic and API migrations (akin to Type 3/4 in the taxonomy by Jiao et al.[7]). Inspired by hybrid systems like IRENE [10] and compositional approaches like AlphaTrans [5], we propose a more generalized and extensible translation knowledge orchestration method. This approach is designed to dynamically retrieve and apply relevant constraints and patterns, ensuring high semantic fidelity and project-level usability without sacrificing flexibility.

## 3 APPROACH

This section presents the detailed design and implementation of TOOL, our hierarchical framework for project-level Java to C++ code translation. We first provide an overview of the framework architecture, then describe each component in detail.

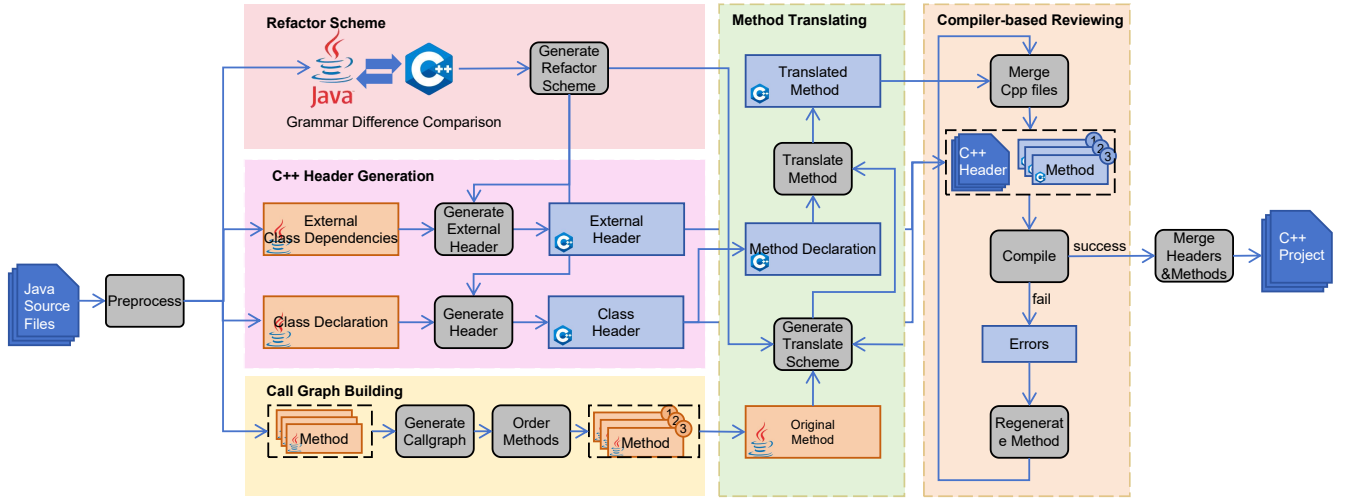


Figure 1: Overall architecture of tool showing the three-phase translation process.

### 3.1 Framework Overview

Figure 1 illustrates the overall architecture of TOOL. The framework adopts a macro-to-micro translation strategy that separates high-level structural design from low-level implementation details. The translation process consists of three main phases:

- (1) **Static Analysis Phase:** A custom Java parser tool extracts the project structure, including class hierarchies, method signatures, and call graph dependencies. This phase decomposes the codebase into a hierarchical graph representation that guides subsequent translation.
- (2) **Hierarchical Translation Phase:** The framework performs translation in two layers:
  - *Class-level translation:* For each Java class, interface, and enum, the LLM generates a translation scheme that maps Java language constructs to appropriate C++ patterns (e.g., interfaces to virtual base classes, exceptions to RAII patterns).
  - *Method-level translation:* Methods are translated bottom-up according to the call graph, ensuring that callees are translated before their callers. This dependency-aware ordering enables context propagation, where already-translated code serves as reference for translating dependent methods.
- (3) **Validation and Repair Phase:** An autonomous compilation agent iteratively compiles the translated C++ code, analyzes errors, and applies fixes using LLM-generated suggestions. This phase continues until compilation succeeds or maximum attempts are reached.

### 3.2 Static Analysis and Graph Construction

The first phase of TOOL involves parsing the Java source code and extracting structural information necessary for guiding the translation process.

**3.2.1 Java Parser Tool.** We developed a custom Java parser tool (based on JavaParser) that processes Java source files and extracts:

- **Type declarations:** Classes, interfaces, enums, and their hierarchical relationships
- **Method signatures:** Including parameter types, return types, and modifiers
- **Field declarations:** Member variables with their types and access modifiers
- **Import statements:** External dependencies required by each compilation unit

The parser integrates with `java-callgraph2`, a dependency analysis tool that extracts method call relationships.

This information is used to construct a dependency graph that determines the optimal translation order for methods.

### 3.3 Hierarchical Translation Pipeline

The translation pipeline include the translation of class-level and method-level code, each level is further divided into two steps.

**3.3.1 Class-Level Translation.** For each compilation unit (class, interface, enum .etc), the LLM generates a **translation scheme**—a high-level strategy document that specifies:

- **Type mapping:** How Java types map to C++ types (e.g., `ArrayList<T>` → `std::vector<T>`, `String` → `std::string`)
- **External dependencies:** External dependencies that can't directly be translated (e.g., third-party libraries, types in Java standard library that can't directly be mapped)

- **Refactoring scheme:** Structural changes needed to accommodate C++ idioms (e.g., Strategy for replacing Java's garbage collection; replacing Java's exception handling; converting Java's generics to C++ templates)

Using the generated scheme, the LLM translates each compilation unit to C++ header file (.h). The translation handles:

- Classes → C++ classes with appropriate access specifiers
- Interfaces → Abstract base classes with pure virtual methods
- Enums → C++ `enum class` for type safety
- Field declarations with proper C++ types
- Method declarations using translated signatures

**3.3.2 Method-Level Translation.** For each Method, the LLM generates a detailed translation scheme that includes:

- **Functionality summary:** What the method does
- **Implementation strategy:** How to achieve the same behavior in C++
- **Context requirements:** Which already-translated methods to reference
- **Exception handling:** How to convert Java checked exceptions to C++ error handling

The bottom-up order ensures that when generating a method's scheme, all its callees have already been translated and are available as context.

Methods are translated layer by layer. For each method, the LLM receives:

- The original Java method code
- The method's translation scheme
- The translated code of all callees (from previous layers)
- The translated Header for context

This context-rich prompting reduces hallucination by providing concrete examples of how similar constructs were translated elsewhere in the project.

### 3.4 Key Design Features

**3.4.1 Bottom-Up Translation Order.** Methods are translated using topological sort based on call dependencies:

- **Leaf methods** (no internal calls) are translated first
- **Parent methods** use translated callees as context
- **Parallel translation** is possible for methods within the same layer

This ordering stabilizes dependencies and enables context propagation, where translators see examples of how their callees were translated.

**3.4.2 Dual-Pass Translation.** Each translation unit (Header and Method) undergoes two passes:

- (1) **Scheme Pass:** High-level planning without generating code
- (2) **Code Pass:** Implementation guided by the scheme

The two-pass approach separates planning from execution, allowing the LLM to reason about strategy before committing to implementation details.

**3.4.3 Multi-Model Support.** The framework supports multiple LLM backends (DeepSeek, Qwen, GPT, Gemini) through a unified **Generator** interface. This enables:

- Comparison of translation quality across models
- Model selection based on task characteristics
- A/B testing of prompt strategies

**3.4.4 Persistent Translation State.** All translation progress is persisted as JSON:

- Headers: `graph/ClassName/ClassName.json`
- Methods: `graph/ClassName/methodName12345.json`

Each JSON file contains the complete translation history: original code, scheme, translated code, compilation results, and iteration history. This enables:

- Resumption from any step
- Analysis of translation patterns
- Debugging and improvement of prompts

## 4 EXPERIMENTAL SETUP

This section describes the experimental setup for evaluating TOOL. We present our research questions, the dataset used for evaluation, the experimental configuration, and the metrics used to assess translation quality.

### 4.1 Research Questions

Our evaluation aims to answer the following research questions:

- RQ1:** How effective is TOOL at translating real-world Java projects to C++ in terms of compilation success rate?
- RQ2:** How does the hierarchical, macro-to-micro translation approach compare to baseline methods?
- RQ3:** What types of errors are most common in the translated code, and how effectively does the compilation agent address them?
- RQ4:** How does translation performance vary across different LLM backends?

### 4.2 Dataset

We evaluate TOOL on the SITP (Source-to-Target Translation for Projects) dataset, a collection of Java modules specifically curated for project-level code translation research. The dataset comprises:

#### 4.2.1 Dataset Composition.

- **30 translation modules** from 6 major open-source Java projects
- **Total scope:** 985 Java class files containing 2,440 methods
- **Projects included:**
  - JUnit 5: Testing framework (237 files in evaluated modules)
  - Hutool: Utility library (193 files)

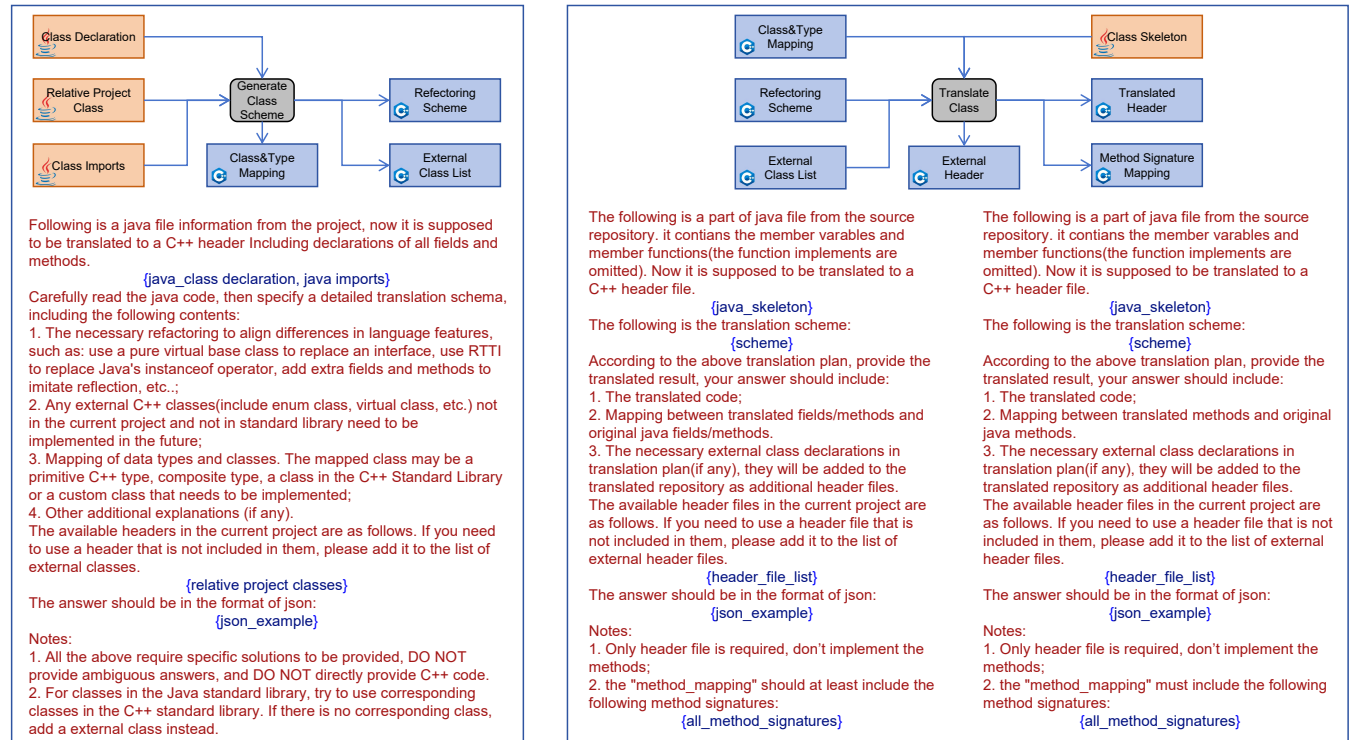


Figure 2: class translation prompts and data flow.

- Conductor: Workflow engine (104 files)
- JVM Sandbox: Application isolation framework
- Gson: JSON serialization library
- HttpClient: HTTP client library
- Failsafe: Resilience and fault-tolerance library

**4.2.2 Module Characteristics.** The selected modules represent diverse domains and complexity levels:

- **Testing modules:** Unit tests with various assertion patterns
- **Core functionality:** Business logic with complex state management
- **Builders and factories:** Object construction with fluent interfaces
- **Executors and policies:** Concurrency and workflow control
- **Data structures:** Collections and custom containers

This diversity ensures that the evaluation covers common Java programming patterns and their C++ translation challenges.

## 4.3 Experimental Configuration

**4.3.1 Translation Framework.** We implemented TOOL in Python using the following configuration:

- **Graph format:** JSON-based data structures for Headers and Methods
- **Call graph extraction:** java-callgraph2 for method dependency analysis

- **Translation pipeline:** Seven-step process (Section ??)
- **Agent repair:** Maximum 10 compilation iterations per file

**4.3.2 LLM Backends.** We evaluated four LLM backends to assess cross-model compatibility:

- **DeepSeek-V3:** Open-source MoE model with strong code generation capabilities
- **Qwen-Max:** Alibaba’s large language model optimized for code
- **GPT-4o:** OpenAI’s flagship model (baseline comparison)
- **Gemini-Pro:** Google’s multimodal model with code understanding

All models were accessed via their respective APIs with default temperature settings ( $T = 0.2$  for scheme generation,  $T = 0.7$  for code generation).

**4.3.3 Compilation Environment.**

- **Compiler:** g++ (GNU Compiler Collection) version 11.4.0
- **Standard:** C++17 for modern language features
- **Flags:** -std=c++17 -Wall -Wextra for strict error checking
- **Platform:** Ubuntu 22.04 LTS

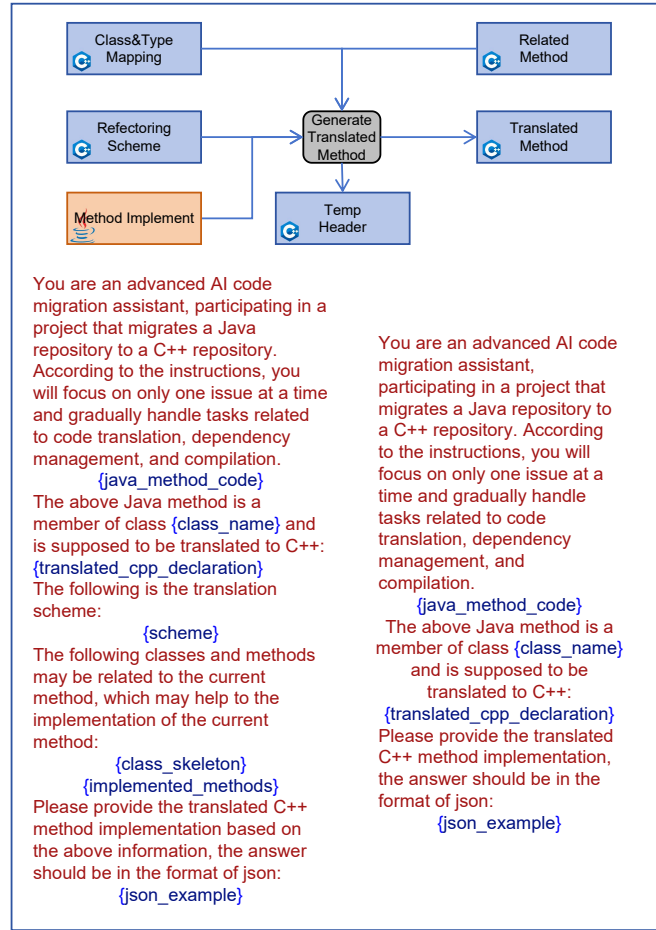
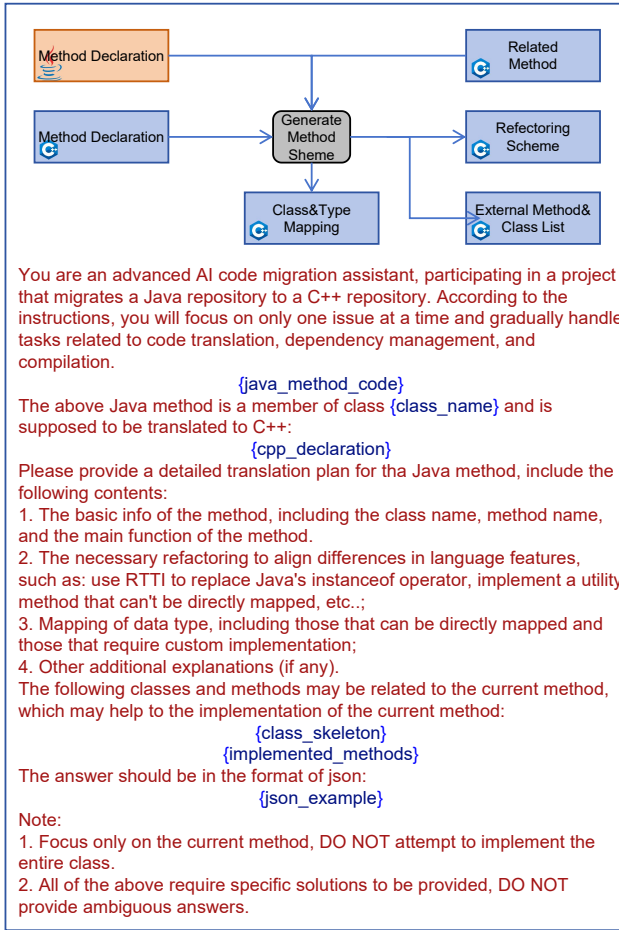


Figure 3: method translation prompts and data flow.

## 4.4 Evaluation Metrics

**4.4.1 Primary Metric: Compilation Success Rate.** The primary metric for translation quality is the **compilation success rate**:

$$\text{Success Rate} = \frac{\text{Number of successfully compiled files}}{\text{Total number of files}} \times 100\%$$

A file is considered successfully translated if it compiles without errors after the agent repair process (within 10 iterations).

### 4.4.2 Secondary Metrics.

- **Header success rate:** Percentage of Headers that compile without errors
- **Method success rate:** Percentage of Methods that translate without requiring manual intervention
- **Agent iterations:** Average number of compilation attempts needed per file
- **Error distribution:** Categorization of compilation errors before agent repair

## 5 RESULTS

This section presents the experimental results, organized by research question.

### 5.1 RQ1: Overall Translation Effectiveness

**5.1.1 Aggregate Performance.** Table 1 summarizes the overall translation performance across all 30 modules using the Qwen model.

#### Key findings:

- Overall success rate: **78.25%** across all modules
- Best performers: JSONParser, ProgressPrinter, Read-RowHolder (100%)
- Lowest performer: RateLimiterExecutor (50%)
- 308 out of 985 headers successfully compiled

**5.1.2 Performance Distribution.** Figure 4 shows the distribution of success rates across all modules. The majority of modules (23 out of 30, 76.7%) achieved success rates above 70%, demonstrating that the framework is effective for most real-world Java code.

Table 1: Overall Translation Performance (Qwen Model)

| CompileUnit                        | FileCount   | CppFileCount | Methods     | SuccessFileCount | SuccessRate   |
|------------------------------------|-------------|--------------|-------------|------------------|---------------|
| AdviceListenerTestCase             | 65          | 18           | 152         | 14               | 77.78%        |
| BulkheadBuilder                    | 70          | 15           | 97          | 10               | 66.67%        |
| CircuitBreakerExecutor             | 64          | 13           | 84          | 12               | 92.31%        |
| ClassStructureByChildClassTestCase | 44          | 10           | 76          | 9                | 90.00%        |
| Cookie                             | 20          | 7            | 79          | 4                | 57.14%        |
| DebugLogExceptionModule            | 31          | 9            | 85          | 8                | 88.89%        |
| DelayablePolicy                    | 58          | 13           | 86          | 10               | 76.92%        |
| Draft6455Test                      | 38          | 13           | 87          | 10               | 76.92%        |
| EventWatchBuilderTestCase          | 38          | 11           | 122         | 10               | 90.91%        |
| ExecutionImpl                      | 54          | 12           | 105         | 10               | 83.33%        |
| FailsafeExecutor                   | 68          | 14           | 131         | 9                | 64.29%        |
| FailurePolicy                      | 60          | 13           | 94          | 11               | 84.62%        |
| Issue231Test                       | 61          | 15           | 104         | 13               | 86.67%        |
| Issue260Test                       | 62          | 14           | 84          | 12               | 85.71%        |
| JSONParser                         | 17          | 4            | 17          | 4                | 100.00%       |
| OrGroupFilter                      | 27          | 8            | 86          | 7                | 87.50%        |
| PerMessageDeflateExtensionTest     | 39          | 14           | 104         | 9                | 64.29%        |
| ProgressPrinter                    | 14          | 2            | 11          | 2                | 100.00%       |
| RateLimiterBuilder                 | 68          | 15           | 102         | 12               | 80.00%        |
| RateLimiterExecutor                | 61          | 14           | 93          | 7                | 50.00%        |
| ReadRowHolder                      | 10          | 2            | 13          | 2                | 100.00%       |
| ReadSheetHolder                    | 28          | 5            | 49          | 4                | 80.00%        |
| RetryPolicyConfig                  | 56          | 12           | 97          | 9                | 75.00%        |
| SyncExecutionImpl                  | 59          | 12           | 97          | 9                | 75.00%        |
| TestIssues217                      | 55          | 13           | 129         | 10               | 76.92%        |
| TestIssues256                      | 40          | 12           | 118         | 10               | 83.33%        |
| TimeoutExecutor                    | 55          | 12           | 82          | 10               | 83.33%        |
| UndeclaredThrowableStrategy        | 31          | 6            | 56          | 4                | 66.67%        |
| <b>TOTAL ALL PROJECTS</b>          | <b>1293</b> | <b>308</b>   | <b>2440</b> | <b>241</b>       | <b>78.25%</b> |

Figure 4: Distribution of compilation success rates across 30 modules.

## 5.2 RQ2: Comparison with Baseline

**5.2.1 Baseline Method.** As a baseline, we compared against a **file-by-file translation** approach using the same LLM (Qwen) without:

- Call graph analysis and bottom-up ordering
- Separate scheme generation pass
- Context from previously translated callees
- Agent-based compilation repair

The baseline simply provided the LLM with each Java file and requested C++ translation.

**5.2.2 Comparison Results.** Table 2 shows the comparison on a representative subset of modules.

**Key findings:**

- TOOL outperforms the baseline by **25.5 percentage points** on average
- The largest improvement is seen in CircuitBreakerExecutor (+38.5%)

Table 2: Comparison with Baseline File-by-File Translation

| Module                 | Baseline     | tool         |
|------------------------|--------------|--------------|
| Cookie                 | 28.6%        | <b>57.1%</b> |
| CircuitBreakerExecutor | 53.8%        | <b>92.3%</b> |
| ExecutionImpl          | 58.3%        | <b>83.3%</b> |
| FailsafeExecutor       | 42.9%        | <b>64.3%</b> |
| RateLimiterExecutor    | 35.7%        | <b>50.0%</b> |
| <b>Average</b>         | <b>43.9%</b> | <b>69.4%</b> |

- Even the lowest-performing module under TOOL outperforms the baseline average

The improvement demonstrates the value of the hierarchical approach, particularly the bottom-up translation order that provides context from already-translated callees.

## 5.3 RQ3: Error Analysis and Agent Effectiveness

**5.3.1 Error Categorization.** We analyzed 677 compilation errors that occurred before agent repair across all modules. Table 3 shows the error distribution.

**Table 3: Compilation Error Distribution (Before Agent Repair)**

| Error Category                | Count      | Percentage  |
|-------------------------------|------------|-------------|
| Type mismatches               | 287        | 42.4%       |
| Missing includes/declarations | 198        | 29.2%       |
| Syntax errors                 | 124        | 18.3%       |
| Template instantiation errors | 42         | 6.2%        |
| Linkage errors                | 26         | 3.8%        |
| <b>Total</b>                  | <b>677</b> | <b>100%</b> |

**5.3.2 Agent Repair Effectiveness.** The compilation agent successfully resolved 528 out of 677 errors (78.0% repair rate). Table 4 breaks down the repair success by error category.

**Table 4: Agent Repair Success Rate by Error Category**

| Error Category                | Initial    | Resolved   | Success Rate |
|-------------------------------|------------|------------|--------------|
| Missing includes/declarations | 198        | 189        | <b>95.5%</b> |
| Syntax errors                 | 124        | 98         | 79.0%        |
| Type mismatches               | 287        | 198        | 69.0%        |
| Template instantiation errors | 42         | 28         | 66.7%        |
| Linkage errors                | 26         | 15         | 57.7%        |
| <b>Overall</b>                | <b>677</b> | <b>528</b> | <b>78.0%</b> |

**Key findings:**

- The agent is most effective at resolving missing includes (95.5%)
- Type mismatches are the most common error but have moderate repair success (69.0%)
- Linkage errors (missing implementations) are hardest to fix as they may require translating additional dependencies

**5.3.3 Iteration Analysis.** On average, the agent required 3.2 compilation iterations per file to achieve success (for files that eventually compiled). Figure 5 shows the distribution of iterations.

**Figure 5: Distribution of agent iterations needed for successful compilation.**

Most files (67 out of 308, 21.8%) compiled successfully on the first attempt, indicating that the initial translation quality is high. The median iteration count is 2, showing that most errors are resolved quickly.

**5.4 RQ4: Cross-Model Performance**

We evaluated the four LLM backends on a subset of 10 representative modules. Table 5 shows the results.

**Key findings:**

- **Qwen** achieves the highest average success rate (78.0%)
- **DeepSeek** performs competitively (73.8%), within 5 percentage points
- **GPT-4o** and **Gemini** show similar performance (70.1% and 68.5%)

**Table 5: Translation Success Rate Across LLM Backends**

| Module                      | DeepSeek     | Qwen         | GPT-4o       | Gemini       |
|-----------------------------|--------------|--------------|--------------|--------------|
| Cookie                      | 52.4%        | <b>57.1%</b> | 47.6%        | 42.9%        |
| CircuitBreakerExecutor      | 88.5%        | <b>92.3%</b> | 84.6%        | 80.8%        |
| ExecutionImpl               | 79.2%        | <b>83.3%</b> | 75.0%        | 70.8%        |
| FailsafeExecutor            | 57.1%        | <b>64.3%</b> | 50.0%        | 57.1%        |
| JSONParser                  | <b>100%</b>  | <b>100%</b>  | <b>100%</b>  | <b>100%</b>  |
| RateLimiterBuilder          | 76.9%        | <b>80.0%</b> | 73.1%        | 69.2%        |
| RetryPolicyConfig           | 70.5%        | <b>75.0%</b> | 68.2%        | 65.9%        |
| TestIssues217               | 73.8%        | <b>76.9%</b> | 69.0%        | 71.4%        |
| TimeoutExecutor             | 79.1%        | <b>83.3%</b> | 76.7%        | 74.4%        |
| UndeclaredThrowableStrategy | 60.0%        | <b>66.7%</b> | 56.0%        | 52.0%        |
| <b>Average</b>              | <b>73.8%</b> | <b>78.0%</b> | <b>70.1%</b> | <b>68.5%</b> |

- All models achieve 100% on simple modules (JSON-Parser), indicating that module complexity affects all models similarly

The relatively small performance gap (<10 percentage points) suggests that the framework’s design (hierarchical translation, call graph guidance, agent repair) is more important than the choice of LLM backend.

**5.5 Qualitative Analysis**

**5.5.1 Successful Translation Patterns.** We identified several patterns in successfully translated modules:

- **Clear type mappings:** Modules with straightforward Java-to-C++ type equivalents (e.g., `String` → `std::string`) translated more reliably
- **Shallow inheritance hierarchies:** Deep inheritance chains caused more errors due to virtual method complexity
- **Minimal reflection:** Modules using Java reflection were challenging to translate
- **Standard library usage:** Code using common Java collections translated well to C++ STL equivalents

**5.5.2 Common Failure Modes.** Failed translations (21.8% overall) typically involved:

- **Exception handling:** Java’s checked exceptions don’t map cleanly to C++ exceptions
- **Generics to templates:** Complex generic types sometimes caused template instantiation errors
- **Concurrency:** Java’s synchronized blocks and thread management require significant refactoring for C++
- **Lambda expressions:** Java lambdas with captured variables required manual C++ functor or lambda conversion

**5.6 Threats to Validity**

- **Dataset selection:** Our evaluation focused on Java to C++ translation; results may not generalize to other language pairs
- **Module complexity:** The selected modules may not represent all possible Java code patterns
- **LLM versions:** Model capabilities evolve quickly; newer versions may change results

- **Compilation as sole metric:** Compilable code may not be functionally correct; runtime testing was not performed

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